

Retained cut kernels, bounded logarithmic defects, and finite-period recovery

Source: `history/EP817_FINITE_PERIOD_20260916/payloads/zeta/workbenches/ep817-finite-period/notes/splitzero-receiving-map.md` **Identity:** `dc18613b9655041ae1618a43cad1e74860d5631989b0797349d641665d496fbd` The Clankers. 16 September 2026.

This receiving note accompanies `finite-period-control.md`, whose complete proofs and standalone verifiers are included beside it. The objects below are the actual finite word sources and value quotients used in the arithmetic construction. The comparison uses the original SplitZero support and receiving-kernel interface; it does not replace the theta source or its infinite quotient by a finite dictionary. No new Lean elaboration or analytic arithmetic-weight theorem is claimed.

1. Two original quotient stages

For a canonical word u , let Ω_u be the actual q -valued generator-choice set, and $X_u = H_q(A_u)$ its distinct numerical image. Retain the free modules

$$U_u = \mathbb{Q}[\Omega_u], \quad V_u = \mathbb{Q}[X_u], \quad q_u(e_\omega) = e_{\Phi_u(\omega)}.$$

Concatenation identifies the actual choice source with Ω_u times Ω_v . After taking the two local numerical quotients, the remaining map is

$$\pi_{u,v} : \mathbb{Q}[X_u \times X_v] \longrightarrow V_{uv}, \quad e_{(x,y)} \longmapsto e_{x+Q(u)y}. \quad (\text{R1})$$

All original radix products $Q(u)$ remain in this formula. The complete source map is $\pi_{\mathbf{u},\mathbf{v}}$ composed with q_u tensor q_v . Both quotient maps are onto, giving

$$0 \longrightarrow \ker(q_u \otimes q_v) \longrightarrow \ker(\pi_{u,v}(q_u \otimes q_v)) \xrightarrow{q_u \otimes q_v} \ker \pi_{u,v} \longrightarrow 0. \quad (\text{R2})$$

For surjectivity at the last term, lift any coefficient vector in $\ker \pi$ by choosing an original word in each of its value fibres. Its full numerical observation vanishes by (R1). The displayed first kernel is exactly the kernel of the restricted map. This proves exactness without identifying the two relation families.

Choose one specified pair p_z in each fibre of (R1). The differences $e_{p_z} - e_{(p_z)}$ for p other than p_z form an integral basis of the receiving kernel. Every original pair and every collision is retained. Over support-indexed subdiagrams, the same map carries a boundary to the zero of its admitted target fibre; it does not send that fibre to the external bottom.

The set bound is independent of the coefficient field. Every fibre of (R1) has at most $C=q-1$ elements because its lower values lie in $[0, C(Q(u)-1)]$ and are congruent modulo $Q(u)$. Hence

$$\dim \ker \pi_{u,v} = F_q(u)F_q(v) - F_q(uv), \quad 1 \leq \kappa(u,v) := \frac{F_q(u)F_q(v)}{F_q(uv)} \leq C. \quad (\text{R3})$$

The quotient size, rather than a generic linear-map norm, controls the multiplicative count loss.

2. Associativity and the complete scalar defect

For three words, both original numerical maps send (x,y,z) to

$$x + Q(u)y + Q(u)Q(v)z.$$

Thus the two parenthesized quotient composites agree on every original basis vector. The receiving-fibre ratios obey the exact identity

$$\boxed{\kappa(u, v)\kappa(uv, w) = \kappa(v, w)\kappa(u, vw)}. \quad (\text{R4})$$

Both sides are $F_q(u)F_q(v)F_q(w)/F_q(uvw)$. With $a(w)=\log F_q(w)$, the logarithmic defect is $a(u)+a(v)-a(uv)$, belongs to $[0, \log C]$, and satisfies the additive version of (R4). The nonzero defect is retained in every iteration; it is not set to zero to obtain a multiplicative model.

Repeated cuts into length- m words bound the total defect by the number of cuts times $\log C$. Dividing by the actual generator reward yields the finite-period horizon error $\log C/(m \cdot n_{\min})$. This is the original arithmetic mechanism behind Theorem FP1. It supplies an a priori finite approximation to an infinite schedule optimization for every specified finite dictionary. The argument is not a claim that the original source quotients become constant at finite length.

A finite safe controller adds an actual bridge between the terminal and initial supports of a candidate path. The bridge has at most $s-1$ edges inside a strongly connected component, but its radices and generator rewards are retained. Its count cost is at most $T^+(s-1)$. Theorem FP2 explicitly carries that factor through the periodic approximation. An active zero-reward reset from earlier work is outside its positive-reward domain; it is not erased or silently assigned a positive reward.

3. The weighted section and the original metric

Give the distinct local values their original word masses $\mu_u(x)$, $\mu_v(y)$. On the pair source the retained mass is

$$w(x, y) = \mu_u(x)\mu_v(y).$$

For a receiving value z , define

$$W_z = \sum_{x+Q(u)y=z} w(x, y).$$

With source Gram $\text{diag}(1/w)$, the exact minimum-norm section is

$$s(e_z) = \sum_{x+Q(u)y=z} \frac{w(x, y)}{W_z} e_{(x,y)}, \quad G_{\text{receiving}} = \text{diag}(1/W_z). \quad (\text{R5})$$

Its image is orthogonal to $\ker \pi$. Indeed for any relation r in one fibre, the inner product with $s(e_z)$ is $\sum(r(x,y))/W_z=0$. Its evaluation is e_z , and its squared norm is $\sum w/W_z^2=1/W_z$. Thus every lift h of e_z has

$$\|h\|^2 = \|h - s(e_z)\|^2 + 1/W_z.$$

The fibre-count bound C limits the number of distinct local value pairs, not their word masses. For a concrete five-admissible example use $A=\{1,2\}$, $b=13$, and $q=5$ at every level. The binary image $\{0,1,2,3\}$ is modular-five-free modulo 13. Its fifth-arity image is $[0,12]$, so all interlevel value fibres are singletons. The local fifth-word multiplicity has maximum three, attained at value four and other central values. Its original m -level maximum multiplicity is exactly 3^m . Therefore the identity from the original numerical quotient metric to unit value mass has norm

$$\boxed{3^{m/2}}.$$

This grows even though every cut in (R1) is injective and has $\kappa=1$. The uniform logarithmic count estimate supplies no unrecorded uniform bound on the original word-to-value metric return.

4. Exact original endpoints in the finite minimizers

The radix-ten dictionary has complete finite minima attained by concatenating the actual macro words BA and BAA. Their generator sets are

$$\{2, 4, 10, 30\}, \quad \{2, 4, 10, 30, 100, 300\},$$

and their original radix products are 100 and 1000. The global scalar map $x \rightarrow x/2$ and its inverse $y \rightarrow 2y$ identify their fifth-images with $[0, 92]$ and $[0, 892]$. It is applied once to the whole numerical image. It does not replace each generator independently or remove any factor from the original radix product.

Every macro concatenation therefore has a digitwise inverse and exact count $93^a 893^b$. In particular the BAA phase has count 893^r at r periods, whereas the original AAB phase has count $2301 * 893^{(r-1)}$. They share their cyclic spectral rate but have different original endpoints. The complete numerical images, actual weights, and the source maps distinguish those finite objects.

5. Scope of the transfer

The present source is a finite arithmetic dictionary or a finite certified controller, with unbounded word length. The new uniform bound controls that infinite schedule variable. The outer family of possible generator blocks and their ranks remains unbounded in the full extremal problem. Neither the scalar defect identity nor the finite-period approximation supplies the missing all-rank arithmetic inequality automatically.

The original SplitZero reconstruction accepts these quotient diagrams and support maps fibrewise. On the analytic theta source, a finite observation can retain an additional kernel beyond the original theta relations. No map in this note asserts that kernel vanishes, nor is an analytic weight estimate attributed to the finite count bounds. The uses of (R1)-(R5) are fully specified finite arithmetic instances of the original quotient interface.

References and execution boundary

The Clankers. (2026, September 16). *Uniform finite-period approximation and exact finite-word switching minima*. Complete provider note and standalone verifiers in this directory. The general proofs use the literal cut map above; the executed finite domains and the independent implementation boundary are listed in the two generated receipts.

KokunoYumeto. (2026). *Reconstruction with changes of support index*, equations D6-D8. `workbenches/splitzero-tandem/tex/support_diagrams.tex` at the previously inspected original source revision `58626a62cd5648e8ae7450fb54d4a4aab2330981`. The original mathematical interface is retained via the delivered predecessor sources; no new formal compilation of that repository is claimed here.

The original anonymous/deleted contributor's four-term construction and sneed-and-feed's Lean upper-bound extension retain their separate attribution.